

A diffuse masa core for Koszmider's non-diagonalizable state

Autonomous proof attempt for arXiv:2002.05230

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Abstract

Koszmider asked whether his ZFC non-diagonalizable pure state on $\mathcal{B}(\ell_2)$ can restrict multiplicatively to a nonatomic masa. We do not settle the weak-closure question. We prove, however, that every pure state arising from the projection face in his construction is multiplicative on a weak-operator dense copy of $C([0, 1])$ inside a nonatomic masa. More precisely, the state's multiplicative domain contains a positive multiplicity-one contraction with essential spectrum $[0, 1]$. We then identify the exact remaining obstruction: continuous endpoint evaluation does not determine a character on diffuse L_∞ .

1 Question and result

Write $H = \bigoplus_{m \geq 1} H_m$ for the finite-dimensional tensor-block decomposition in [1]. Its projection family has the form

$$P_\alpha = \bigoplus_{m \geq 1} P_{\alpha, m}, \quad \alpha \in 2^{\mathbb{N}},$$

where $P_{\alpha, m}$ is rank one on the tensor coordinate indexed by $\alpha|m$ and is the identity on all other coordinates. If $\alpha|m \neq \beta|m$, then $P_{\alpha, m}$ and $P_{\beta, m}$ act on distinct tensor factors and commute. The construction chooses a pure state ϕ from the face

$$\mathcal{F} = \{\rho : \rho \text{ is a state on } \mathcal{B}(H), \quad \rho(P_\alpha) = 1 \text{ for all } \alpha \in 2^{\mathbb{N}}\}.$$

It is non-diagonalizable, hence its restriction to no atomic masa is multiplicative. The source asks whether it can be multiplicative on a nonatomic masa [1, p. 2].

Theorem 1 (Diffuse C^* -core). *Let ϕ be any pure state in $\mathcal{F}$. There are a positive contraction $A \in \mathcal{B}(H)$ and a nonatomic masa $\mathcal{M} \subset \mathcal{B}(H)$ such that*

$$\sigma(A) = \sigma_e(A) = [0, 1], \quad \mathcal{M} = W^*(A), \quad \phi(f(A)) = f(1) \quad (f \in C([0, 1])).$$

In particular, $C^(A) \cong C([0, 1])$ is weak-operator dense in $\mathcal{M}$ and ϕ is multiplicative on $C^*(A)$.*

Thus the source state admits a diffuse masa on which all continuous moments already form a character. The theorem stops strictly short of asserting that $\phi|_{\mathcal{M}}$ is pure.

2 A commuting Boolean family in the multiplicative domain

For a state ρ , let

$$\text{MD}(\rho) = \{x : \rho(x^*x) = |\rho(x)|^2 = \rho(xx^*)\}.$$

This is a unital C^* -subalgebra, and $\rho(xy) = \rho(x)\rho(y)$ whenever $x \in \text{MD}(\rho)$ and $y \in \mathcal{B}(H)$.

Lemma 2. $\mathcal{K}(H) \subset \text{MD}(\phi)$ and $P_\alpha \in \text{MD}(\phi)$ for every α .

Proof. A pure state of $\mathcal{B}(H)$ which does not annihilate $\mathcal{K}(H)$ is a vector state: its irreducible GNS representation restricts nontrivially to the compacts, whose unique nonzero irreducible representation is the identity representation. A vector state is diagonalizable, whereas ϕ is not. Hence $\phi|_{\mathcal{K}(H)} = 0$. For compact k this gives $\phi(k^*k) = \phi(kk^*) = 0$, so $k \in \text{MD}(\phi)$. Finally, $\phi(P_\alpha) = \phi(P_\alpha^2) = 1$, which puts every P_α in the same domain. $\square$

Choose distinct sequences $\alpha_1, \alpha_2, \dots \in 2^{\mathbb{N}}$. Inductively choose increasing integers N_n so that

$$\alpha_n|_m \neq \alpha_j|_m \quad (m \geq N_n, j < n),$$

and truncate only finitely many blocks:

$$E_n = 0 \oplus \dots \oplus 0 \oplus \bigoplus_{m \geq N_n} P_{\alpha_n, m}.$$

Then $E_n - P_{\alpha_n}$ has finite rank. Lemma 2 gives $E_n \in \text{MD}(\phi)$ and $\phi(E_n) = 1$.

Lemma 3. *The projections E_n commute. Moreover, for finite disjoint sets $F, G \subset \mathbb{N}$, the signed Boolean word*

$$R_{F,G} = \prod_{n \in F} E_n \prod_{n \in G} (I - E_n)$$

has infinite rank.

Proof. For $i < j$, the j th projection is zero before block N_j ; after that block the two source projections occupy distinct tensor coordinates. Thus they commute everywhere.

Fix finite disjoint F, G . On every sufficiently late block, all relevant binary prefixes are distinct. The compression of $R_{F,G}$ to that block is the tensor product of rank-one projections for indices in F , nonzero complements of rank-one projections for indices in G , and identities on unused coordinates. It is nonzero. This happens on infinitely many orthogonal blocks, so $R_{F,G}$ has infinite rank. $\square$

Set

$$D = \sum_{n=1}^{\infty} 2^{-n} E_n.$$

The series converges in norm inside $\text{MD}(\phi)$ and $\phi(D) = 1$. Lemma 3 says that the essential joint spectrum of (E_n) is the full binary cube: every finite cylinder has a nonzero image in the Calkin algebra. Applying the binary expansion map gives

$$\sigma(D) = \sigma_e(D) = [0, 1]. \tag{1}$$

(Equivalently, every dyadic cylinder has infinite-dimensional spectral range, and their mesh tends to zero.)

3 From an atomic model to a diffuse generator

Let T be multiplication by t on $L_2([0, 1])$. It is a positive multiplicity-one contraction and

$$\sigma(T) = \sigma_e(T) = [0, 1], \quad W^*(T) = L_\infty([0, 1]),$$

the standard nonatomic masa. We use the following classical form of the Weyl–von Neumann theorem: two bounded self-adjoint operators with the same essential spectrum and the same multiplicities at

isolated spectral points off that spectrum are unitarily equivalent modulo a compact operator [2]. Here there are no isolated points off the essential spectrum. By (1), some unitary $U : L_2([0, 1]) \rightarrow H$ and compact K obey

$$A := UTU^* = D + K.$$

Lemma 2 and the C^* -algebra property of the multiplicative domain give $A \in \text{MD}(\phi)$. Therefore

$$\phi(A) = \phi(D) + \phi(K) = 1, \quad \phi(f(A)) = f(1) \quad (f \in C([0, 1])).$$

Finally $W^*(A) = UL_\infty([0, 1])U^*$ is a nonatomic masa and $C^*(A)$ is weak-operator dense in it. This proves Theorem 1.

4 Why the full question remains

Let $e_\varepsilon = 1_{[0, 1-\varepsilon]}(A) \in W^*(A)$. Since $I - A \geq \varepsilon e_\varepsilon$ and $\phi(I - A) = 0$, positivity yields

$$\phi(e_\varepsilon) = 0, \quad \phi(1_{(1-\varepsilon, 1]}(A)) = 1 \quad (\varepsilon > 0).$$

Thus $\phi|_{W^*(A)}$ is concentrated in every neighborhood of the endpoint 1. But this does not force a character on diffuse L_∞ : endpoint evaluation on $C([0, 1])$ has many character extensions to L_∞ , and a nontrivial convex combination of two such extensions has the same values on all continuous functions and on all endpoint neighborhoods while failing to be multiplicative.

The missing step is therefore not spectral or compact-perturbative. It is a complete Boolean-lattice statement about all measurable spectral projections. Continuous functional calculus survives modulo compacts; internal spectral projections are discontinuous at points of $[0, 1] = \sigma_e(A)$ and are not controlled by Weyl–von Neumann equivalence. This is the precise gap between the theorem above and the source’s nonatomic-masa question.

References

- [1] P. Koszmider, *A non-diagonalizable pure state*, Proc. Natl. Acad. Sci. USA **117** (2020), 33084–33089; arXiv:2002.05230v3.
- [2] I. D. Berg, *An extension of the Weyl–von Neumann theorem to normal operators*, Trans. Amer. Math. Soc. **160** (1971), 365–371.